

User Manual

R-GPT (R-Gene Prediction Tool) is a web-based platform designed for the identification and classification of plant disease resistance (R) proteins. These proteins play a critical role in plant immune responses against pathogens such as fungi, bacteria, and viruses. R-GPT integrates multiple machine learning models to provide a multi-stage prediction pipeline, enabling users to:

- Identify R-gene sequences
- Determine classification feasibility
- Assign sequences to specific R-gene classes
- Evaluate prediction confidence using probability scores

1. Input File Requirements

R-GPT accepts protein sequences in FASTA format, where each sequence must begin with a header line starting with ">".

Example:

>Protein_1

MKTLLAILSVLSAFSVQA...

Input Constraints:

- Only standard amino acids are allowed: ACDEFGHIKLMNPQRSTVWY
- Sequence length must be between 50 and 5000 amino acids
- Multiple sequences can be submitted in a single input file

2. How to Submit a Job

To perform prediction using R-GPT:

1. Navigate to the Server page
2. Paste protein sequence(s) OR upload a FASTA file
3. Click on the Submit button
4. The server will process the input sequences
5. You will be redirected to the results page



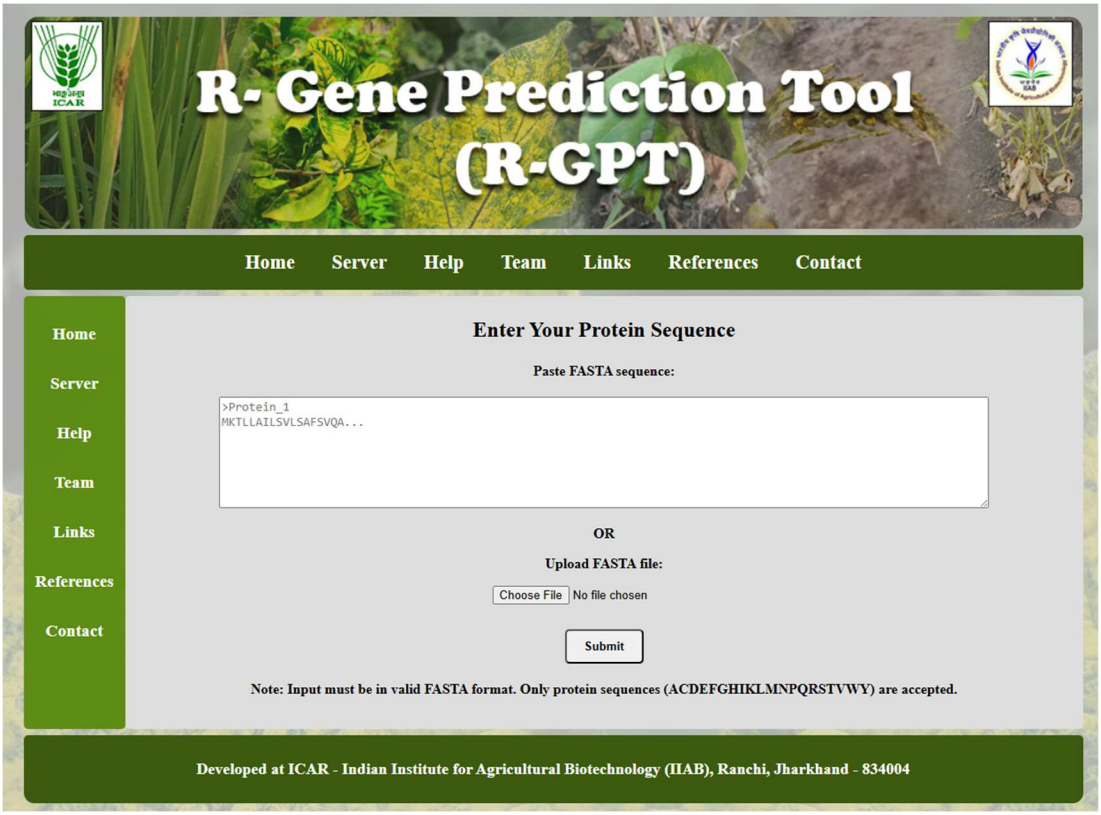


Figure 1: Server Page

3. Prediction Workflow

R-GPT follows a three-stage prediction pipeline:

Model 1: R-gene Identification

- Performs binary classification
- Determines whether a sequence is an R-gene or not

Model 2: Classifiability Assessment

- Evaluates whether the identified R-gene can be reliably classified
- Filters sequences that cannot be assigned to known classes

Model 3: R-gene Classification

Assigns sequences to specific R-gene classes which includes: Kinase-Kinase, LRR-TrD, LRR-TrD-Kinase, Lysin, NBS-LRR-CC, NBS-LRR-TIR, TrD-CC.



4. Output Description

The output consists of three CSV files, corresponding to each prediction stage.

➢ Model 1 Output (model1_predictions.csv)

This file contains binary classification results.
Columns:

a. Seq_ID: Input sequence identifier
b. Predicted_Label:
   1 → R-gene
   0 → Non R-gene
c. Probability_R_gene: Confidence score of prediction

Higher probability indicates stronger confidence in R-gene prediction.

	Seq_ID	Predicted_Label	Probability_R_gene
2	pdb|7XE0|A	1	0.9998896
3	AGP75918.1	1	0.99989426
4	KAE8803279.1	1	0.99968374
5	VAH68925.1	1	0.9996426
6	KAF7040043.1	1	0.9998373
7	KAI5002253.1	1	0.9991767
8	KAE8815593.1	1	0.99917847
9	KQJ87241.1	1	0.9977155
10	KAK1647754.1	1	0.9990336
11	KAF7069690.1	1	0.9997571
12	KAF7059857.1	1	0.9996137
13	CAM0908593.1	1	0.99946994
14	AAG42168.1	1	0.9994216
15	XBJ21643.1	1	0.9990946
16	KAF6986911.1	1	0.99891376
17	KAI5009866.1	1	0.9986243
18	XBI96846.1	1	0.9939375
19	KAF7012523.1	1	0.9974037
20	KAF7012526.1	1	0.99896276
21	KAK1619430.1	1	0.9998336
22	KAF7019951.1	1	0.98268145

Figure 2: Model 1 prediction output



➢ **Model 2 Output (model2_predictions.csv)**

This file evaluates whether a predicted R-gene is classifiable.
Columns:

1. Predicted_Label:
    a. 1 → Classifiable
    b. 0 → Not classifiable
2. Probability_Classified: Confidence score

Sequences marked as non-classifiable will not proceed to meaningful class assignment.

1	Seq_ID	Predicted_Label	Probability_Classified
2	pdb|7XE0|A	1	0.998
3	AGP75918.1	1	0.996
4	KAE8803279.1	1	0.962
5	VAH68925.1	1	1
6	KAF7040043.1	1	0.964
7	KAI5002253.1	1	0.95
8	KAE8815593.1	1	0.96
9	KQJ87241.1	1	0.956
10	KAK1647754.1	1	0.992
11	KAF7069690.1	1	0.954
12	KAF7059857.1	1	0.99
13	CAM0908593.1	1	0.99
14	AAG42168.1	1	1
15	XBJ21643.1	1	0.99
16	KAF6986911.1	1	0.9
17	KAI5009866.1	1	0.99
18	XBI96846.1	1	0.992
19	KAF7012523.1	1	0.982

Figure 3: Model 2 classification output



- **Model 3 Output (model3_predictions.csv)**

This file provides detailed classification into specific R-gene classes.
Columns:
Predicted_Class: Final assigned class
Class Probabilities: Probability scores for each class
The class with the highest probability is selected as the final prediction.

Example interpretation:
If Prob_CLASS=NBS-LRR-CC = 0.99, the sequence strongly belongs to the class NBS-LRR-CC.

	Seq_ID	Predicted_Class	Prob_CLASS=Kinase-Kinase	Prob_CLASS=LRR-TrD	Prob_CLASS=LRR-TrD-Kinase	Prob_CLASS=Lysin	Prob_CLASS=NBS-LRR-CC	Prob_CLASS=NBS-LRR-TIR	Prob_CLASS=TrD-CC
2	pdb|7XE0|A	CLASS=NBS-LRR-CC	0.000134187	0.000141717	0.000177201	0.000103209	0.99905634	0.000285195	0.000102056
3	AGP75918.1	CLASS=NBS-LRR-CC	0.000141062	0.000141716	0.0001772	0.000103209	0.99904966	0.000285193	0.000102055
4	KAE8803279.1	CLASS=NBS-LRR-CC	8.37E-05	0.000144499	0.000160013	0.000115523	0.9990206	0.000361673	0.000114231
5	VAH68925.1	CLASS=NBS-LRR-CC	5.17E-05	7.92E-05	0.000145183	7.55E-05	0.9993774	0.000196248	7.46E-05
6	KAF7040043.1	CLASS=NBS-LRR-CC	4.34E-05	6.26E-05	9.48E-05	6.49E-05	0.99941653	0.000253584	6.41E-05
7	KAI5002253.1	CLASS=NBS-LRR-CC	0.000313674	0.000304419	0.00076583	0.000338332	0.99736416	0.000625241	0.000288176
8	KAE8815593.1	CLASS=NBS-LRR-CC	0.001374664	0.001779922	0.00216537	0.001964843	0.98429996	0.006253604	0.002161433
9	KQJ87241.1	CLASS=NBS-LRR-CC	6.20E-05	0.000237468	0.000117771	0.00011375	0.9929251	0.006423581	0.000120466
10	KAK1647754.1	CLASS=NBS-LRR-CC	0.000261087	0.00076718	0.000984358	0.000375761	0.9910802	0.005696394	0.000834906
11	KAF7069690.1	CLASS=NBS-LRR-CC	0.001531337	0.000430154	0.000512618	0.000430178	0.9818336	0.014907518	0.000354492
12	KAF7059857.1	CLASS=NBS-LRR-CC	9.75E-06	3.19E-05	3.02E-05	1.73E-05	0.9998269	7.00E-05	1.41E-05
13	CAM0908593.1	CLASS=NBS-LRR-CC	6.75E-05	6.98E-05	4.31E-05	7.33E-05	0.99893725	0.00075177	5.75E-05
14	AAG42168.1	CLASS=NBS-LRR-CC	0.0004216	0.000148694	0.000167794	0.000140671	0.99828815	0.000733277	9.99E-05
15	XBJ21643.1	CLASS=NBS-LRR-CC	0.00017299	7.74E-05	0.000148816	6.59E-05	0.9981288	0.001345997	6.00E-05
16	KAF6986911.1	CLASS=NBS-LRR-CC	0.000324785	0.000195679	0.000156668	0.000222788	0.9984605	0.000492493	0.00014678
17	KAI5009866.1	CLASS=NBS-LRR-CC	2.15E-05	3.96E-05	2.31E-05	2.69E-05	0.9996387	0.000226179	2.43E-05
18	XBI96846.1	CLASS=NBS-LRR-CC	0.000188976	0.000146934	0.000102432	0.000125243	0.9985601	0.000762431	0.000113988
19	KAF7012523.1	CLASS=NBS-LRR-CC	0.000128127	0.00017578	5.64E-05	0.000138538	0.9986205	0.000760871	0.000119774
20	KAF7012526.1	CLASS=NBS-LRR-CC	0.000118241	0.000177371	6.41E-05	0.000139793	0.9985776	0.000802152	0.000120858

Figure 4: Model 3 class prediction output

- **Probability Distribution Chart**

For each sequence, an interactive probability distribution chart is generated.
Interpretation:
Each segment represents a predicted class. The segment size corresponds to probability.
Higher percentage = stronger prediction confidence



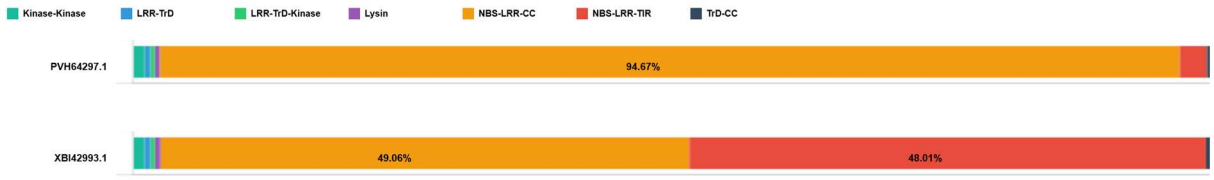


Figure 5: Probability distribution chart

5. Downloading Results

Users can download:

- ZIP file containing:
  - Input FASTA
  - Model predictions (CSV files)
- Probability distribution chart image

6. Important Notes

- Invalid or improperly formatted sequences will not be processed.
- Ensure correct FASTA formatting before submission.
- Processing time depends on the number of sequences.
- Results are stored for 7 days on the server.

7. Contact

For queries, suggestions, or technical support, please contact the development team via the Contact page of the web server.